Bhuvana Annamreddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.07
Sri Chaitanya Junior college	Class XII (BIEAP)	2021	97.00%
Sri Chaitanya Techno School	Class X (SSC)	2019	10.00%

SCHOLASTIC ACHIEVEMENTS

Selected as **Scientist/Engineer 'SC', ISRO** -- All India Rank **5**, ICRB 2025

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant, IIT Hyderabad** -- Assisted four courses: Signal Processing (Prof. K. Sri Rama Murthy), Differential Equations & Transform Techniques (Dr. Lakshmi Prasad), Random Processes and Error Correcting Codes (Dr. Anjana A. Mahesh).
- **Class Representative, JNTUK** -- Represented the Department of ECE for the academic year 2022-23.

SKILLS

Languages: Verilog HDL, C, MATLAB

EDA Tools: Xilinx Vivado, Intel Quartus Prime

Soft Skills: Time Management, Problem Solving, Critical Thinking

RELEVANT COURSES

Digital: Digital IC Design, Introduction to VLSI Design, Memory Circuits, Digital Electronics, Computer Organization & Architecture

Communication & Signal Processing: Principles of Digital Communications, Wireless Communications, Advanced Digital Signal Processing, DSP Lab

Machine Learning: Pattern Recognition & Machine Learning, Random Variables & Stochastic Processes

PROJECTS

RTL Design of 32-bit RISC-V Processor | *Verilog, Icarus Verilog, GTKWave*

Single-Cycle Architecture

- Designed and implemented a 32-bit RISC-V (RV32I) single-cycle processor supporting R, I, S, B and J-type instructions.
- Built the datapath using a program counter, instruction and data memory, register file, ALU, immediate generator and control unit.
- Verified functionality using test programs covering arithmetic, memory and branch instructions.

Pipelined Architecture

- Developed a 5-stage pipelined version (IF/ID/EX/MEM/WB) by adding pipeline registers between stages.
- Implemented a hazard unit handling data hazards by forwarding, load-use hazards by stalling, and control hazards by flushing.
- Conducted functional verification using a testbench with programs that trigger all three hazard types.

8×8 Signed Booth Multiplier -- Pipelined RTL Design | *Verilog, Icarus Verilog, GTKWave*

- Designed an 8×8 signed multiplier using radix-4 Booth encoding, reducing partial products from 8 to 4.
- Developed a 5-stage pipelined version by adding registers after partial-product generation and inside each adder of a two-level adder tree.
- Conducted functional verification using a testbench covering signed operand combinations across positive, negative and zero inputs.

RTL Design of FIFO Buffers | *Verilog, Quartus*

Synchronous FIFO

- Designed a parameterised synchronous FIFO with configurable data width and depth, using read and write pointers within a single clock domain.
- Implemented full and empty flag generation using an occupancy counter, blocking writes when full and reads when empty to prevent overflow and underflow.

Dual-Clock Asynchronous FIFO

- Designed a dual-clock asynchronous FIFO for safe data transfer between independent write and read clock domains.
- Implemented Gray-coded pointers passed through two-flop synchronisers to avoid metastability during clock domain crossing, with an extra MSB to distinguish full from empty at the same address.

UART Transmitter and Receiver -- RTL Design | *Verilog, Vivado*

- Designed a UART transmitter and receiver operating at a fixed baud rate, using FSMs to handle start bit, 8-bit data and stop bit framing.
- Integrated both into a top module and verified functionality using a testbench.

EXTRACURRICULAR

Hobbies: Badminton, Arts & Crafts, Watching Movies

CERTIFICATES

- Digital Design using Verilog -- NPTEL
- VLSI Design -- Udemy
- Digital Timing Basics -- Udemy

MTECH THESIS

Unequal Error Protection for Function-Correcting Partition Codes | *Under the guidance of Dr. Anjana A. Mahesh*

- Formulated a coding framework in which each block of a function-induced message partition is assigned its own error-correction capability, unifying classical unequal error protection with function-correcting codes.
- Derived and proved matching lower and upper bounds on optimal redundancy, and designed systematic encoding constructions attaining the optimum.
- Developed layered constructions that incrementally raise protection levels by appending parity layers to an existing code.